

SUMMARY

Lead AI/ML System Architect specializing in the end-to-end lifecycle of multi-modal models. Skilled in constructing robust fine-tuning pipelines and architecting highly scalable, low-latency deployment infrastructure. Adept at bridging the gap between advanced AI research and enterprise-grade production systems with a focus on reliability and performance.

EDUCATION

MS, CS, Arizona State University | GPA 3.92 *August 2017 - December 2019*

BTech, CS, Maulana Abul Kalam Azad University of Technology | GPA 8.67/10 *July 2012 - July 2016*

TECHNICAL SKILLS & COURSEWORK

Languages: Python, GOLANG, C++, SQL, Bootstrap
ML Technologies: Pytorch, Huggingface, vLLM, Numpy, TensorFlow, OpenCV, Pandas, Scikit-learn, Matplotlib
Cloud DevOps: Docker, Kubernetes, Helm, AWS
Databases: MySQL, DynamoDB, OCI NoSQL, PostgreSQL

WORK EXPERIENCE

AI/ML Lead, Full-Stack Backend Developer | A10Networks - Incedo | Pune *June 2023*

- Leading the R&D team at A10Networks to drive the company's AI vision by creating innovative solutions for building and deploying LLM guardrails to safeguard applications against Cyber attacks.
 - Designed and engineered end-to-end pipelines for fine-tuning multi-modal models, seamlessly transitioning advanced vision-and-text architectures from research phases into production-ready assets.
 - Spearheaded the architecture and deployment of scalable infrastructure optimized for multi-modal solutions, ensuring low-latency inference, high availability, and robust system reliability under heavy production loads.
 - Architected a high-performance data retrieval and request-management system, implementing an enhanced Golang rate-limiting solution that reduced CPU usage by 45%, memory by 38%, and improved response times by 70%, while enabling the targeted filtering and retrieval of 10K attacked IPs from a 65M-entry data warehouse in under 13 seconds.
- Technology Stack:** PyTorch, HuggingFace, vLLM, GOLANG, FastAPI, DynamoDB, Docker, Helm, Kubernetes

Research Engineer, Lead | DiDi Labs | California *Nov 2021 - April 2023*

- Architected & Implemented a Graph-Based Neural Network to forecast the heading direction of pedestrians within a scene to aid our car's decision-making process.
 - Boosted the efficiency of the model from 65% to 72%.
 - Optimize and deploy models using techniques such as model compression, pruning, and quantization to improve performance and reduce computational cost.
 - Constructed different metrics to measure the accuracy of our algorithms.
- Technology Stack:** C++, Python, PyTorch

Backend Software Developer | AmazonGO | Washington *Feb 2020 - Sept 2021*

- Designed Java APIs based on SQS to receive requests from clients and utilize vision-based algorithms for predicting the ultimate shopping events.
 - Collaborate with data scientists and software engineers to take machine learning models from research and development to production.
- Technology Stack:** Python, Java, Numpy, Scikit-learn, OpenCV, SQS, Guice

OPEN SOURCE PROJECTS

GPTs from Scratch | [Github](#)

- Built a GPT-style model in PyTorch from scratch inspired by Llama and Mistral.
- Codebase designed for modularity and extensibility to support experimentation and learning principles.

LLMs for Everyone – Fully Offline | [Github](#)

- Developed an open-source platform that enables easy offline use of Hugging Face models with integrated back-end and front-end.
- Simplified running LLMs locally without the internet - no cloud or API dependency.
- Delivered a seamless LLM chat and completions playground.